

Food Technology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15 and 17

Total marks – 100

Section I Pages 2–6

20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II Pages 9–18

50 marks

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III Page 19

15 marks

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV Page 20

15 marks

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Which of the following materials is most suitable for use in food-processing equipment?
- (A) Glass
 - (B) Plastic
 - (C) Aluminium
 - (D) Stainless steel
- 2** A company produces a copy of an existing food product.
- What type of food product development is this?
- (A) Me too
 - (B) Innovation
 - (C) New to world
 - (D) Line extension
- 3** A crop is produced without the use of synthetic chemicals.
- This means it has been produced by
- (A) mass production.
 - (B) organic farming.
 - (C) industrial agriculture.
 - (D) commercial cropping.
- 4** What is the best temperature for storing frozen goods?
- (A) 0°C
 - (B) –4°C
 - (C) –18°C
 - (D) –100°C

- 5** Which of the following best describes a prototype?
- (A) The final product
 - (B) The marketing plan
 - (C) A model of a product produced for testing
 - (D) A record of costs associated with a product
- 6** Vegetarians need to take particular care to include which of the following in their diet?
- (A) Iron
 - (B) Sodium
 - (C) Vitamin D
 - (D) Dietary fibre
- 7** El Niño is a climate pattern that causes extreme weather conditions.
- Which sector of the Australian food industry is directly affected by an El Niño cycle?
- (A) Food retail
 - (B) Agriculture and fisheries
 - (C) Food service and catering
 - (D) Food processing/manufacturing
- 8** Milk is heated to 72°C for 15 seconds before packaging.
- What is this process known as?
- (A) Irradiation
 - (B) Fermentation
 - (C) Pasteurisation
 - (D) High pressure processing

9 A muffin-making chain produces a low-GI muffin.

Which driver of food product development is the company most likely responding to?

- (A) Technological developments
- (B) Consumer demand for convenience
- (C) Increase in single parent households
- (D) Market concerns such as dietary considerations

10 For which of the following groups is folate recommended as a supplement?

- (A) Athletes preparing for events
- (B) Women planning a pregnancy
- (C) Children who live in apartments
- (D) Elderly people with sedentary lifestyles

11 What is the main role of a design brief in food product development?

- (A) To plan profit margins for the company
- (B) To provide a possible solution to a problem
- (C) To gather information about competitors' products
- (D) To provide an outline of the purpose of the development

12 A food product contains the following ingredients:

soybean oil, whole eggs, water, sugar, white vinegar, vegetable gum (405), food acid (citric), natural colour (carrot extract), flavour, emulsifier (lecithin).

What is this product most likely to be?

- (A) Ice-cream
- (B) Margarine
- (C) Carrot cake
- (D) Mayonnaise

- 13** Which of the following most affects working conditions in the food service and catering sector of the Australian food industry?
- (A) Penalty rates
 - (B) Climate change
 - (C) Labelling legislation
 - (D) Self-service check-outs
- 14** Which of the following organisations oversees approvals to produce genetically modified foods?
- (A) National Food Authority
 - (B) Food and Grocery Council
 - (C) Food Standards Australia and New Zealand
 - (D) National Association for Sustainable Agriculture Australia
- 15** A retailer offers a prize as part of a marketing strategy. When customers try to claim the prize they are told it is not available.
- Which piece of legislation is being breached?
- (A) *Food Act 2003* (NSW)
 - (B) *Food Regulation Act 2015* (NSW)
 - (C) *Work Health and Safety Act 2011* (Cth)
 - (D) *Competition and Consumer Act 2010* (Cth)
- 16** A large burger chain is trialling a process where customers can design their own burgers.
- Which of the following are internal factors that are most likely to have contributed to this trial?
- (A) Company image and financial position
 - (B) Personnel expertise and market concerns
 - (C) Ecological and technological environments
 - (D) Financial position and economic environments

- 17** Which of the following occurs when the body produces antibodies to destroy a food protein?
- (A) Food poisoning
 - (B) A diabetic reaction
 - (C) A food intolerance
 - (D) An allergic reaction
- 18** Which of the following is an example of a probiotic?
- (A) Anthocyanin
 - (B) Cellulose
 - (C) Lactobacillus
 - (D) Lignin
- 19** Which of the following ingredients, used in commercial bread-making, is required by law to be fortified with iodine?
- (A) Salt
 - (B) Flour
 - (C) Butter
 - (D) Plant oils
- 20** In which of the following preservation processes is an atomiser most likely to be used?
- (A) Freezing
 - (B) Sous vide
 - (C) Dehydration
 - (D) Fermentation

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Centre Number

Section II

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50 marks**Attempt Questions 21–26**

Student Number

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (5 marks)

- (a) Identify ONE quality management system used in food production. **1**

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- (b) Describe how the system identified in part (a) assists in the manufacture of foods that are safe for public consumption. **4**

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Question 22 (5 marks)

An experiment was conducted to investigate food spoilage.

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The experiment took place over five days. Four equal sized slices of apple were placed in individual dishes as follows:

Dish 1 – 1 slice of apple

Dish 2 – 1 slice of apple dipped in lemon juice

Dish 3 – 1 slice of apple covered with plastic film

Dish 4 – 1 slice of apple which had been blanched.

The dishes were left at room temperature.

Explain the expected results.

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Centre Number

Section II (continued)

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Student Number

Question 23 (12 marks)

A multinational chocolate company is looking to improve its profitability. The company decides to develop a chocolate product which uses new technologies including edible packaging.

- (a) Describe the activities that may be conducted at the idea generation and screening stage of developing the chocolate product. **3**

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- (b) Complete the following SWOT analysis in relation to the development of the chocolate product. **4**

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Question 23 continues on page 12

Question 23 (continued)

- (c) Explain marketing strategies for the new chocolate product that the company could use.

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End of Question 23

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Centre Number

Section II (continued)

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Student Number

Question 24 (7 marks)

- (a) Complete the following table.

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<i>Level of operation</i>	<i>Example</i>
	Jams sold at local markets
Multinational	
	Local pizza shop
Large company	

- (b) Discuss the impact of gender issues on employment in the various sectors of the Australian food industry. Use specific examples to support your answer.

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Centre Number

Section II (continued)

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Student Number

Question 25 (9 marks)

Australia has recently signed free trade agreements with a number of countries.

- (a) Discuss the possible effects of changes to trade policy on the Australian food industry. **4**

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Question 25 continues on page 16

Question 25 (continued)

- (b) Describe storage conditions and distribution systems required to maintain maximum quality in fresh fruit and vegetables for export.

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End of Question 25

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Section II (continued)

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Student Number

Question 26 (12 marks)

An adolescent regularly consumes the following daily diet:

- toasted muesli with full cream milk and sugar
- meat pie with tomato sauce
- chocolate donut
- lasagne with a green salad and garlic bread
- ice-cream with chocolate chips.

- (a) Identify a diet-related disorder that may develop if this type of diet were to be continued into adulthood. 1

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- (b) For the diet-related disorder identified in part (a) explain:

- (i) the physical effects. 3

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Question 26 continues on page 18

Question 26 (continued)

- (ii) the economic costs.

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- (c) Describe how nutritional health can be promoted through the use of social media. Use specific examples to support your answer.

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End of Question 26

Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Question 27 (15 marks)

A shelf-stable high fibre drinking yoghurt, that does not require refrigeration, has been developed by food manufacturers.

- | | | |
|-----|---|----------|
| (a) | Justify a target market for this product. | 2 |
| (b) | Explain how the active non-nutrients in this product benefit health. | 5 |
| (c) | Identify the main features of a package suitable for this product, including materials to be used. Explain how these features may contribute to the success of the product. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question in a SEPARATE writing booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Explain how sectors of the Australian food industry have responded to lifestyle changes and environmental concerns.

End of paper